THE UNIVERSITY OF AZAD JAMMU & KASHMIR, MUZAFFARABAD

COURSE TITLE	CALD
COURSE CODE	CS-1205
LAB TITLE	SR LATCH
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DEPARTMENT OF SOFTWARE ENGINEERING

CALD

Lab 9

Objective:

This experiment aims to demonstrate the working of an SR latch using both the NOR and NAND Gates.

S-R Latch

- The **S-R latch** is the **simplest type of sequential logic circuit**.
- It is a **bistable multivibrator,** meaning it has **two stable states**: storing either 0 or 1.
- Unlike combinational circuits, its output depends on **both present input and past output** (i.e., it has memory).

- An S-R latch is a **basic memory element**.

Inputs and Outputs

- Inputs: **R (Reset)** and **S (Set)**
- Outputs: **Q** and **$\bar{Q}$** (complement of Q)

Basic Implementations

There are **two common ways** to build an S-R latch:

(A) Using NOR gates

- Inputs are active **HIGH**.
- Circuit: Two cross-coupled NOR gates.

Truth Table (NOR-based latch):

S	R	Q (next state)	$\bar{Q}$
0	0	No change (holds state)	Previous
0	1	0	1
1	0	1	0
1	1	**Invalid** (both outputs = 0)	

(B) Using NAND gates

- Inputs are active **LOW**.
- Circuit: Two cross-coupled NAND gates.

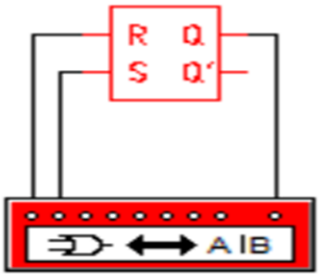
Truth Table (NAND-based latch):

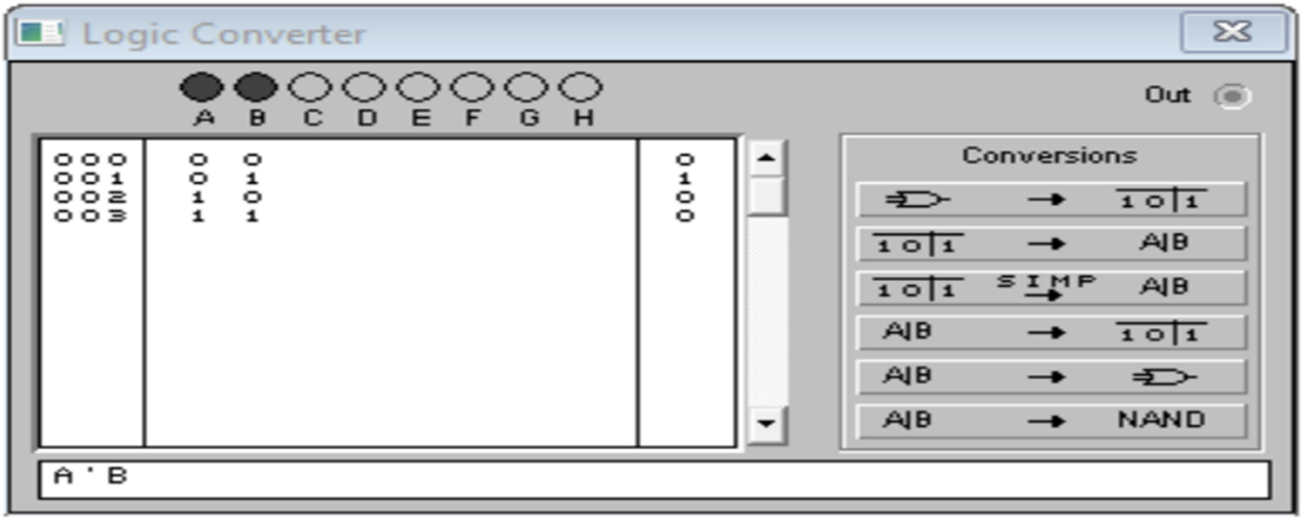
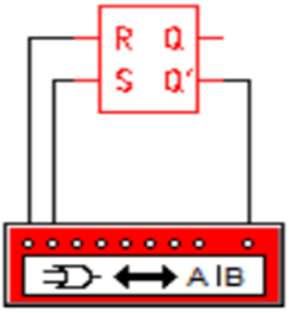
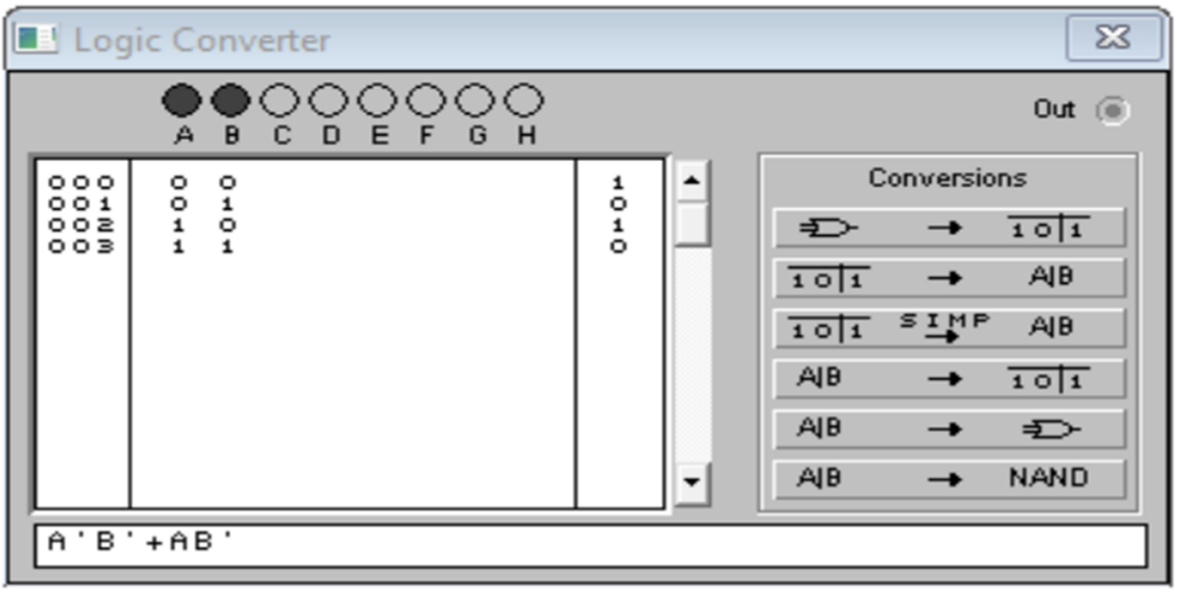
S	R	Q (next state)	$\bar{Q}$
1	1	No change (holds state)	Previous
1	0	0	1
0	1	1	0
0	0	**Invalid** (both outputs = 1)	

Behavior Explanation

- **Set (S=1, R=0 in NOR / S=0, R=1 in NAND)** → Forces Q = 1.
- **Reset (S=0, R=1 in NOR / S=1, R=0 in NAND)** → Forces Q = 0.
- **Hold (S=R=0 in NOR / S=R=1 in NAND)** → Keeps previous state.
- **Invalid (S=R=1 in NOR / S=R=0 in NAND)** → Produces undefined output (Q and $\bar{Q}$ both same).

Truth table & Equation (Q)





Truth table & Equation (Q`)